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Sandro Lüscher
luescher@zda.uzh.ch

University of Zurich

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Seat Reversals and Candidate Status Changes under Biproportional Representation

Sandro Lüscher[1*]

[1*]Centre for Democracy Studies Aarau (ZDA), University of Zurich, Villa Blumenhalde, Küttigerstrasse 21, 5000, Aarau, Switzerland, ORCID: 0000-0002-6092-5422 .

Corresponding author. E-mail: luescher@zda.uzh.ch;

Abstract

Biproportional representation improves aggregate party proportionality by constraining district-level allocations through canton-wide party entitlements. This article examines how those constraints affect candidate outcomes in a personalized list system. It distinguishes classic seat reversals from the broader set of party–district entitlement changes generated by biproportional allocation. When a party's district entitlement changes while candidate rankings remain fixed, the electoral threshold moves within the local list.

The article analyses the 2023 Zurich cantonal election through an exact mechanical counterfactual comparing the observed biproportional seat matrix with independent district-level Webster allocation. District magnitudes, party votes, the legal quorum, candidate votes, and candidate rankings are held constant. Fourteen of 188 party–district cells differ, representing seven seat transfers and fourteen candidate status changes. Seven of the 180 parliamentary seats, or 3.9 percent, would have had a different occupant. Two transfers produce discordant seat allocations in which a locally weaker party receives more seats than a stronger party. Biproportionality reduces the Gallagher index from 2.64 to 1.73 and the Loosemore–Hanby index from 4.81 to 2.55. Six transfers correct aggregate party-seat totals, while one reflects additional territorial redistribution. Vote gaps across the shifted candidate thresholds vary substantially, showing that mechanically equivalent status changes can cross either narrowly or clearly differentiated local rankings. The findings reveal a measurable trade-off between aggregate proportionality and local allocation continuity.

Keywords: Biproportional representation, seat reversals, discordant seat allocations, candidate status changes, personalized proportional representation, Switzerland

JEL Classification: C63 , D71 , D72

1 Introduction

Proportional representation (PR) is usually evaluated at the party level. The central question is whether parties receive legislative seat shares that correspond to their electoral support (Rae 1967; Taagepera and Shugart 1989). Yet PR is rarely implemented as a single polity-wide conversion of votes into seats. Territorial districts divide the allocation problem into local contests, and variation in district magnitude and the geographic distribution of party support can produce aggregate disproportionality even when every district uses a proportional formula (Cox 1997; Gallagher and Mitchell 2005).

Biproportional representation addresses this problem by combining two constraints. Parties receive aggregate seat entitlements based on their support across the entire electoral territory, while districts retain fixed magnitudes. The final allocation is a party-by-district seat matrix whose party totals and district totals satisfy both requirements (Pukelsheim 2017). This design can improve aggregate proportionality without abolishing territorial districts or changing the ballot structure. It has therefore attracted particular interest in fragmented electoral systems with unequal district magnitudes (Lüscher 2024).

The mathematical literature on biproportional apportionment recognizes that this aggregate correction can have locally counterintuitive consequences. Maier et al. (2010, p. 375) describe *discordant seat allocations*: within a district, a party with more votes may receive fewer seats than a locally weaker party because the allocation must simultaneously satisfy aggregate party entitlements and fixed district magnitudes. Such cases can be understood as *seat reversals*, because the ordering of parties by local votes is reversed in their ordering by seats. They constitute a particularly visible local price of global proportionality (Demange 2011).

Seat reversals do not exhaust the local consequences of biproportionality. A party's district entitlement may differ from what independent district PR would produce even when the resulting seat allocation preserves the local ordering of parties by votes. This broader category of party–district entitlement changes matters especially in personalized list systems.Through cumulation, panachage, and the striking of names, voters can alter candidates' preference-vote totals and thereby their ranking within the local party list (Lutz et al. 2018). Candidate votes determine who occupies the relevant ranks, but those ranks yield mandates only up to the number of seats assigned to the party in the district. A globally induced change in the party's district entitlement therefore changes how far down the locally determined ranking election extends.

This article connects these two stages of the allocation process. It asks how often biproportionality changes party–district seat entitlements relative to independent district PR, when those changes produce classic seat reversals, and how they affect the personal composition of parliament. The central mechanism is sequential. Canton-wide party constraints first alter some district seat entitlements. Those changes then move the electoral threshold within the affected local party lists. Candidate votes do not determine whether the party–district cell changes, but they determine which individual occupies the rank crossed by the shifted threshold.

The vote gap across that threshold is therefore not a second cause of the outcome. It is a descriptive characteristic of the candidate-level consequence. A small gap indicates that the allocation change separates two similarly supported adjacent candidates; a large gap indicates that it crosses a clearly differentiated point in the ranking. This distinction matters for substantive and normative interpretation, but the candidate status change itself follows from the altered party entitlement.

This formulation also defines the limits of the argument. The article does not claim that biproportionality generally makes candidate votes ineffective or that it directly reduces electoral accountability. Candidate votes continue to determine the order in which local party mandates are filled. Nor does every allocation-induced candidate change involve either a classic seat reversal or a knife-edge candidate contest. The object of study is the local incidence of global allocation constraints and the continuity of candidate outcomes. Broader accountability and legitimacy implications remain conditional on how voters and candidates understand and respond to the system.

The empirical application is the 2023 election to the Zurich cantonal parliament. The design compares the official biproportional outcome with a district-based PR counterfactual. Seats are allocated independently within each district using Webster/Sainte-Laguë, while district magnitudes, party votes, the Zurich list-group quorum, candidate votes, and candidate rankings are held fixed. The comparison isolates the immediate mechanical consequence of removing the canton-wide upper-apportionment constraint. It does not estimate how parties, candidates, or voters would behave under a different electoral system.

The results reveal a modest but clearly measurable trade-off. Fourteen of 188 party–district cells differ between the two seat matrices. These changes amount to seven seat transfers, alter the status of fourteen candidates, and place different candidates in seven of the 180 parliamentary seats. Two of the seven transfers generate pairwise seat reversals in the observed biproportional allocation, whereas no reversal occurs under independent district PR. Four of the fourteen candidate status changes are directly associated with these discordant allocations. The remaining ten demonstrate that the candidate-level incidence of global proportionality is broader than the established seat-reversal concept. At the same time, biproportionality improves aggregate proportionality: the Gallagher index declines from 2.64 to 1.73 and the Loosemore–Hanby index from 4.81 to 2.55.

The article makes four contributions. First, it extends the literature on discordant seat allocations from party-level outcomes to the personal composition of parliament. Second, it distinguishes three related but non-equivalent phenomena: classic seat reversals, cross-rule changes in party–district entitlements, and allocation-induced candidate status changes. Third, it identifies the sequential mechanism linking global party constraints to local candidate outcomes and uses the candidate-threshold vote gap to characterize whether a status change crosses a narrowly or clearly differentiated ranking boundary. Fourth, it measures the exchange between aggregate proportionality and local candidate continuity and benchmarks both seat transfers and seat reversals under perturbations of the Zurich vote matrix.

The remainder of the article proceeds as follows. Section 2 develops the conceptual framework and relates seat reversals to candidate-level outcomes. Section 3 describes the Zurich system and illustrates the allocation mechanism. Section 4 sets out the counterfactual design, measures, and perturbation analysis. Section 5 reports the empirical results. Section 6 presents the perturbation benchmark and robustness checks. Section 7 discusses the implications and limitations, and Section 8 concludes.

2 Global Proportionality and Local Candidate Outcomes

2.1 Two Levels of Electoral Choice

Personalized proportional representation combines two electoral decisions. Voters influence how many seats a party receives and which candidates occupy those seats. These decisions are analytically distinct but institutionally connected. Party votes determine the number of mandates available to a party, while candidate votes determine the ordering of candidates within the party list.

Under independent district PR, both decisions are embedded in the same territorial unit. Party votes cast in a district determine the party's local seat entitlement, and candidate votes cast for the party's local candidates determine who fills those seats. The local result can therefore be reconstructed from the votes and rankings within that district.

Biproportional representation modifies the first part of this relationship. Candidate rankings remain locally determined, but the number of seats available to each party in a district is constrained by the party's canton-wide entitlement. A party may consequently obtain one more or one fewer seat in a district than it would under an allocation based exclusively on that district's votes. Local candidate votes still determine the ranking, but they are applied to a seat entitlement that is no longer fully determined locally.

This does not make the outcome arbitrary or disconnected from votes. The allocation remains rule-based and reproducible. It does, however, change the territorial structure of the vote–seat relationship. The election of a local candidate may depend not only on votes cast in that candidate's district but also on the distribution of party support across the canton.

2.2 Local Allocation Autonomy

The relevant institutional distinction can be described as one of *local allocation autonomy*. A district possesses greater allocation autonomy when its own party votes determine the complete distribution of its seats. Under independent district PR, each district settles its party-seat allocation separately. The resulting canton-wide party totals are simply the sum of these local outcomes.

Under biproportionality, districts retain a fixed number of seats, but they no longer determine independently which parties receive them. The final allocation must simultaneously ensure that each district retains its prescribed number of seats and each party receives its canton-wide entitlement. The seat allocated to a party in one district is therefore connected to allocations elsewhere. A local party-seat outcome may be adjusted because the aggregate party result produced by independent district allocation would otherwise be incompatible with canton-wide proportionality.

This concept is narrower than electoral accountability. Accountability additionally concerns whether voters understand electoral consequences, whether candidates anticipate electoral sanctions, and whether representatives respond to voters (Powell 2000). The present study does not directly observe these behavioral processes. It examines the institutional structure that precedes them: the extent to which local candidate outcomes are determined by local rather than non-local allocation constraints.

2.3 Discordant Seat Allocations and Seat Reversals

The most visible departure from local allocation autonomy is a discordant seat allocation. Consider two parties competing in the same district. If the first party receives more local votes but fewer seats than the second, the ordering of the parties by votes and seats is reversed. Following the apportionment literature, this article refers to such a pairwise discordance as a *seat reversal* (Maier et al. 2010; Demange 2011).

Seat reversals are a within-allocation concept. They compare two parties in the same district under a single electoral rule. They should be distinguished from a cross-rule change in a party's district entitlement. A party–district cell changes when the party receives a different number of seats under biproportionality than under independent district PR. Such a change may produce a seat reversal, but it need not do so. A locally weaker party may gain a seat while still receiving fewer seats than the stronger party, or a stronger party may lose a seat while retaining a larger delegation.

This distinction is central to the candidate-level extension. The established seat-reversal concept identifies particularly stark party-level discordance. Candidate outcomes, however, change whenever a party's district entitlement changes, regardless of whether the final allocation reverses the local ordering of parties. Seat reversals are therefore a subset of the broader local incidence of globally constrained allocation.

2.4 From Party-Seat Entitlements to Candidate Thresholds

The candidate-level mechanism is sequential. First, the allocation rule determines how many seats party p receives in district d. Second, the party's locally determined candidate ranking fills those seats from the top down. If the party receives three seats, ranks one through three are elected and the electoral threshold lies between ranks three and four. If the entitlement changes to two or four seats, the threshold moves accordingly.

Biproportionality therefore operates directly on the party–district seat entitlement. It does not reorder candidates. Instead, it changes the number of positions in the existing ranking that result in election. Candidate votes determine who occupies those positions, but they do not determine whether the global matrix constraint adds or removes a party seat in the district.

Two descriptive measures help characterize the resulting change. The first is the proximity of the affected party–district cell to the relevant district Webster seat boundary. It indicates how close the independent district allocation was to assigning a different number of seats. The second is the *candidate-threshold margin*: the vote gap between the candidates on either side of the rank boundary moved by the allocation change. This gap indicates whether the threshold crosses a narrowly or clearly differentiated point in the local ranking.

These measures have different roles. Local quotient proximity describes the party-seat allocation boundary. The candidate-threshold margin describes the candidate-level incidence after that boundary has moved. Neither is treated as an independent cause of the matrix change. In particular, a narrow candidate vote gap does not make the party more likely to gain or lose a district seat under biproportionality.

A special case occurs when a party moves between zero and one seat. One allocation rule then provides the party with local representation while the other does not. This is highly consequential for the party and its electorate, but there is no interior last-elected/first-non-elected boundary under the zero-seat allocation. Such cases are therefore reported separately rather than assigned an artificial candidate-threshold margin.

2.5 From Party-Seat Changes to Candidate Status Changes

A comparison of independent district PR and biproportionality produces two party-by-district seat allocations. Where both allocations assign the same number of seats to a party, the elected candidates are identical because candidate rankings are held constant.

Where the entitlements differ, candidate status changes follow under three conditions: the ranking is the same under both allocations, mandates are assigned in rank order, and the list contains enough candidates to fill the larger entitlement. If a party receives one additional seat under biproportionality, the candidate occupying the next rank becomes elected. If the party receives one fewer seat, the candidate occupying the final elected rank under the more generous allocation is no longer elected. A change of more than one seat affects the corresponding number of consecutive ranks.

Every transferred parliamentary seat consequently has two candidate-level effects: one candidate gains elected status and one candidate loses elected status elsewhere in the matrix. Seven transferred seats imply fourteen affected candidates but only seven seats occupied by a different person. Counting status changes captures the number of individuals affected; counting differently occupied seats captures the change in the composition of parliament.

The number of candidate status changes is therefore not an outcome that needs to be explained through candidate-level regression once the two seat matrices are known. Candidate-level analysis instead addresses who is affected, whether the affected transfer coincides with a seat reversal, and whether the shifted threshold crosses a narrow or wide vote gap. The formal relationship between seat-matrix differences, transferred seats, and candidate status changes is presented in Appendix A.

2.6 Aggregate Party Correction and Territorial Allocation

Not all differences between the two seat allocations have the same analytical meaning. Some transfers are required because independent district PR produces canton-wide party totals that differ from the proportional party entitlements. Parties that receive too many seats in the sum of the district allocations must lose seats, while parties that receive too few must gain them.

These corrections must always be implemented in particular districts. They are therefore territorial in practice, even though their minimum number is determined by differences in aggregate party totals. A further territorial rearrangement may be necessary after those totals have been corrected. This occurs when the corrected party seats cannot be distributed across districts while simultaneously preserving all district magnitudes without additional changes. Such a transfer leaves the canton-wide seat total of the parties involved unchanged and alters only where their seats are located.

The distinction is thus between transfers required to reconcile aggregate party totals and additional territorial rearrangements that leave those totals unchanged. All transfers remain consequences of biproportionality and all are implemented through party–district changes. The distinction identifies which changes can already be inferred from aggregate party correction and which arise only from the two-dimensional distribution of seats.

2.7 Analytical Expectations

The framework yields three expectations.

First, globally constrained allocation may generate classic seat reversals, but not every party–district change should produce one. Seat reversals are a particularly strong form of local discordance rather than a complete measure of local adjustment.

Second, each difference in a party's district entitlement should produce a corresponding candidate status difference, provided that rankings are fixed and lists have sufficient capacity. This is a mechanical implication of the electoral rules rather than a probabilistic hypothesis.

Third, the candidate-level incidence of these changes should vary. Some status changes should occur in districts with seat reversals and others without them. The shifted candidate threshold may cross a narrow vote gap, a clearly differentiated point in the ranking, or no interior boundary at all because the party moves between zero and one seat. These are descriptive differences in the character and potential salience of the status change, not competing explanations of why the party–district entitlement changed.

3 Institutional Setting and Allocation Logic

3.1 Personalized Biproportional Representation in Zurich

The Zurich cantonal parliament has 180 seats distributed across 18 electoral districts. District magnitudes are fixed, but aggregate party entitlements are determined canton-wide and subsequently distributed across districts through biproportional apportionment. Candidate mandates are then assigned within each party–district list according to candidate votes.

Eligibility is determined at the list-group level. A list group qualifies for allocation if it obtains at least 5 percent of the vote in one district or at least 3 percent canton-wide.

This distinction is important. Applying the threshold independently to every party–district cell would exclude locally weaker cells belonging to an otherwise eligible list group and would generate a different counterfactual. The primary analysis therefore implements the legal group-level rule. A cell-wise interpretation is retained only as a diagnostic robustness specification.

Zurich is an informative case because it combines unequal district magnitudes, multiple parties, personalized candidate voting, and an established biproportional procedure. It also permits an exact empirical reconstruction: official list-level and candidate-level results provide the party votes, observed party–district seat matrix, candidate votes, rankings, and elected status required for the counterfactual analysis.

3.2 Independent District PR and Biproportionality

Under independent district PR, each district allocates its seats separately. For district d, party votes determine s_{pd}^{D} through Webster/Sainte-Laguë. The district totals are respected, but the sum of these local allocations need not reproduce the party seat totals implied by canton-wide support.

Under biproportionality, party entitlements are first determined at the canton-wide level. A second allocation distributes these seats across districts while preserving every district magnitude (Pukelsheim 2017; Maier et al. 2010; Serafini and Simeone 2012). Candidate rankings are not part of this matrix calculation. They are applied only after the party–district seat entitlements have been established.

Figure 1 summarizes the difference.

3.3 Stylized Example

Consider two districts with magnitudes $m_A = 6$ and $m_B = 4$ and two parties, X and Y. Party X receives 800 votes in District A and 160 in District B; party Y receives 150 and 890 votes, respectively. Independent district Webster allocation yields $(X, Y) = (5, 1)$ in District A and $(1, 3)$ in District B, giving aggregate totals of six seats for X and four for Y. Canton-wide support, however, implies five seats for each party. A biproportional matrix satisfying both party and district totals is $(4, 2)$ in District A and $(1, 3)$ in District B.

The two matrices differ by one lost seat for X and one gained seat for Y in District A. Their L_1 distance is two, corresponding to one transferred seat and two candidate status changes. If the local rankings are $X_1, \ldots, X_5$ and Y_1, Y_2, candidate X_5 is elected only under district PR and candidate Y_2 only under biproportionality. No candidate overtakes another. The effective candidate threshold moves because the party-seat threshold changes.

The example also shows the limited but useful role of the candidate-threshold margin. Whether the relevant candidates are separated by one vote or several hundred votes has no effect on the number of status changes generated by the matrix difference. The gap only characterizes whether the shifted threshold crosses a narrowly or clearly differentiated point in the local ranking.

Importantly, the example does not produce a seat reversal. Party X remains locally stronger than party Y in both votes and seats. The candidate-level effect therefore

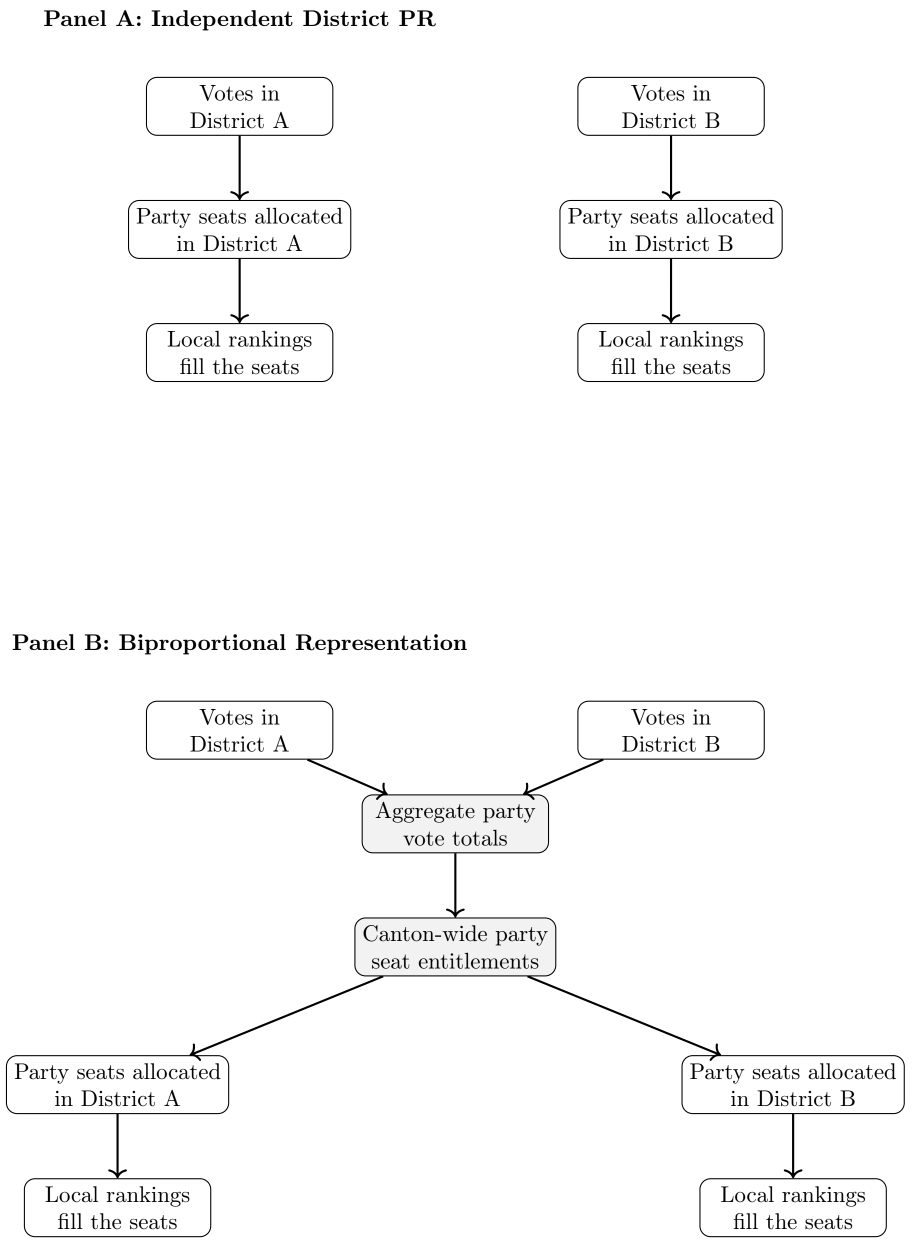

Fig. 1 Vote–seat mappings under independent district proportional representation and biproportional representation. Candidate rankings remain local under both rules, but the party–district seat entitlement is locally autonomous only under independent district PR.

arises from a changed party entitlement even though the local ordering of the parties remains concordant. This illustrates why seat reversals capture only a subset of allocation-induced candidate changes.

4 Research Design and Measurement

4.1 A Mechanical Counterfactual

The analysis compares two allocations of the same 2023 Zurich election data. The observed allocation is the official biproportional result. The counterfactual allocates seats independently within each district using Webster/Sainte-Laguë. Both allocations use the same district magnitudes, party and list-group votes, party–district contest structure, legal list-group eligibility rule, candidate votes, candidate rankings, and number of candidates on each list.

The only changed element is the constraint linking local allocations to canton-wide party entitlements. This is a static mechanical counterfactual. It estimates the immediate allocation effect of removing the upper-apportionment constraint, not the total effect of replacing the electoral system. Parties might nominate differently, voters might behave differently, and campaigns might adapt under an actual district-PR regime. Those behavioural effects are deliberately held outside the estimand.

Webster is the primary counterfactual because it preserves the divisor logic of the observed system while removing its two-dimensional constraint. This minimises the number of institutional dimensions changed simultaneously. D'Hondt and Hare allocations are reported as sensitivity checks rather than equally plausible substitutes for the primary counterfactual.

4.2 Data Preparation and Validation

The analysis uses three official resources published by the Office for Statistics and Data of the Canton of Zurich: canton-wide list results, party–district list results, and candidate-level district results (Amt für Statistik und Daten des Kantons Zürich 2023). Candidate rankings are taken from the official rank variable when it uniquely orders all candidates on a list. Where that variable is incomplete or non-unique, rankings are reconstructed from candidate votes using deterministic tie-breaking information.

Several validation steps precede the substantive analysis. Observed and counterfactual party–district seats must sum to every district magnitude. The observed party–district matrix is independently recreated using a biproportional allocation routine. Official elected status is compared with rank-based status. Finally, the relationship between cell changes and candidate status changes is verified for every party–district list. All eleven core diagnostics pass: the official matrix is recreated exactly, official and rank-based candidate outcomes coincide, and all lists contain sufficient candidates.

4.3 Core Estimands

The analysis reports six groups of quantities.

First, *cell-level change* is the number and share of party–district cells whose seat entitlement differs between biproportionality and independent district PR.

Second, *matrix distance* records the total number of cell-level seat differences and the corresponding number of transferred seats. The formal relationship is set out in Appendix A.

Third, *candidate continuity* is measured by the number of candidates whose status differs and by the number and share of parliamentary seats occupied by a different candidate. The latter is more substantively informative than dividing status changes by the full candidate pool, because most candidates are not competitive for a mandate under either rule.

Fourth, *discordant seat allocation* is measured within each allocation. A pairwise seat reversal occurs when one eligible party has more district votes but fewer seats than another eligible party. The analysis reports the number of districts with at least one reversal, the number of discordant party pairs, and the number of districts in which the locally strongest party is overtaken. It also identifies whether each of the seven observed transfer pairs directly produces a seat reversal.

Fifth, *aggregate proportionality* is measured using the Gallagher index and the Loosemore–Hanby index, two established but differently constructed measures of seat–vote disproportionality (Gallagher 1991; Loosemore and Hanby 1971; Taagepera and Grofman 2003), together with the maximum absolute percentage-point deviation between party vote and seat shares. The trade-off is expressed as the number of transfers and candidate status changes associated with the observed reduction in disproportionality.

Sixth, the decomposition in Appendix A separates transfers required to correct aggregate party totals from additional territorial redistribution within the biproportional matrix.

4.4 Measuring Seat Reversals

Seat reversals are identified through pairwise comparisons of eligible parties within each district. Vote ties and seat ties are not classified as reversals. For each discordant pair, the analysis records the locally stronger and weaker parties, their vote shares and seat totals, the vote difference, and whether the locally strongest party in the district is overtaken.

This measure differs from the counterfactual comparison. A seat reversal can be identified from one observed allocation without specifying an alternative electoral rule. By contrast, a party–district cell change is defined by comparing the same party and district across two rules. The empirical connection between the concepts is assessed by asking how many transfer districts contain a reversal, how many gain–loss pairs are themselves discordant, and how many candidate status changes occur in those districts.

4.5 Local Seat Proximity and Candidate-Threshold Margins

Local seat proximity is measured from the independent district Webster allocation. For each party–district cell, the analysis calculates the relative distance to gaining the next district seat and the relative buffer protecting the final awarded seat. Their minimum provides a quotient-closeness measure. This measure describes the local allocation boundary against which the biproportional result is compared. It is used descriptively because the seat-matrix changes are jointly determined and a single election offers too few independent events for credible explanatory modelling.

The candidate-level consequence is characterized by the vote gap across the rank boundary moved by the allocation change. For lists with a conventional interior threshold, the *candidate-threshold margin* is the difference between the candidate-vote totals immediately above and below that boundary. The analysis reports the raw gap and a normalized value obtained by dividing it by mean candidate votes on the party–district list. The normalized value permits comparison across lists with different vote scales; it is not a percentage-point difference in candidate vote shares.

Where a list moves between zero and one seat, no interior threshold exists under the zero-seat allocation. These cases are classified separately. Candidate-vote dispersion and the Gini coefficient are retained as contextual measures of within-list differentiation.

Neither local quotient proximity nor the candidate-threshold margin is used as a causal predictor of status change. The matrix determines whether the party gains or loses a district seat. The threshold margin only indicates whether the resulting candidate status change crosses a near-tie or a clearly differentiated point in the voter-determined ranking.

4.6 Dirichlet Perturbation Sensitivity Analysis

To assess whether the Zurich results depend narrowly on the exact 2023 vote totals, the analysis uses a Dirichlet-based local sensitivity simulation. Probabilistic simulations have previously been used to evaluate electoral systems in empirically defined environments and to generate district-based election outcomes (Chamberlin and Featherston 1986; Mitra 2021). More directly, Cohen and Hanretty (2024) show how realistic vectors of party vote and seat shares can be generated with the Dirichlet distribution. The present analysis adapts this approach to perturb the observed party shares separately within each Zurich district.

Let $\mathbf{p}_d$ denote the observed vector of party vote shares in district d. For each simulation r, a perturbed vote-share vector is drawn as

$$\widetilde{\mathbf{p}}_d^{(r)} \sim \mathrm{Dirichlet}\left(\kappa \mathbf{p}_d\right), \tag{1}$$

where κ controls the concentration of the draws around the observed shares. The expected value remains $\mathbf{p}_d$; lower values of κ produce broader perturbations, whereas higher values produce outcomes closer to the observed election. The analysis uses $\kappa \in \{40, 80, 160\}$ to represent progressively narrower perturbations. These values are calibration choices rather than estimates of electoral uncertainty.

District magnitudes and the observed party contest structure are held fixed. The simulated shares are rescaled to the observed number of voter equivalents in each district, and the Zurich list-group eligibility rule is applied to each simulated vote matrix. Each election is then allocated independently by district using Webster and biproportionally using the same procedure as in the main analysis. Each concentration condition contains 1,000 successfully allocated simulations.

The simulations record changed party–district cells, seat transfers, candidate status changes implied by matrix distance, discordant districts, pairwise seat reversals, and changes in aggregate disproportionality. The resulting distributions provide a local sensitivity benchmark for Zurich 2023, not a sampling distribution, electoral forecast, or estimate of outcomes in other electoral settings.

5 Results

5.1 The Proportionality–Continuity Trade-off

Table 1 reports the principal estimands. The observed and counterfactual matrices contain 188 party–district cells. Fourteen cells, or 7.4 percent, differ between the allocation rules. The affected cells are distributed across seven districts and eight parties. In every affected district, one party gains and one party loses a seat relative to independent district PR.

The distance between the two seat matrices is fourteen cell-level changes, corresponding to seven seat transfers. Because candidate rankings are fixed and all lists have sufficient capacity, these differences translate directly into fourteen candidate status changes: seven candidates gain and seven lose elected status under biproportionality. Equivalently, seven of the 180 parliamentary seats would have been occupied by a different candidate under independent district Webster allocation. This is 3.9 percent of the legislature.

Table 1 Core estimands and aggregate proportionality, Zurich 2023

Quantity	Value
Parliamentary seats	180
Party–district cells	188
Changed party–district cells	14
Share of cells changed	7.4%
Affected districts	7
Affected parties	8
L_1 distance between seat matrices	14
Seat transfers	7
Candidate status changes	14
Candidates gaining status under biproportionality	7
Candidates losing status under biproportionality	7
Seats occupied by a different candidate	7
Share of parliamentary seats with a different occupant	3.9%
Candidate-change rate among all candidates	0.83%
Candidate-change rate among threshold-adjacent candidates	4.53%
Gallagher index, independent district PR	2.64
Gallagher index, biproportionality	1.73
Loosemore–Hanby index, independent district PR	4.81
Loosemore–Hanby index, biproportionality	2.55

Table 2 reports the fourteen changed party–district cells as seven paired seat transfers. In every affected district, one party gains and one party loses exactly one seat under biproportionality relative to independent district PR. The result is therefore not driven by a single large reallocation but by several geographically dispersed adjustments.

Table 2 Changed party–district seat allocations under biproportionality

District	Party	District PR	Biproportional	Difference
Zurich IV	AL	0	1	+1
	Greens	2	1	−1
Zurich VI	AL	0	1	+1
	FDP	2	1	−1
Hinwil	FDP	1	2	+1
	SP	2	1	−1
Uster	GLP	2	3	+1
	Greens	2	1	−1
Winterthur-Land	EVP	0	1	+1
	Greens	1	0	−1
Andelfingen	The Centre	0	1	+1
	SVP	2	1	−1
Dielsdorf	FDP	1	2	+1
	SVP	5	4	−1

Note: District PR refers to independent district-level Webster allocation using the Zurich list-group quorum. Difference is calculated as biproportional seats minus district-PR seats. Each district contains one one-seat gain and one corresponding one-seat loss.

Every changed party–district entitlement differs by exactly one seat. The fourteen changed cells therefore form seven paired transfers distributed across seven districts rather than a concentrated reallocation in a single part of the seat matrix.

Biproportionality reduces the Gallagher index by 0.91 points and the Loosemore–Hanby index by 2.25 points. Expressed as a descriptive exchange rate, the observed adjustment entails 7.70 seat transfers, or 15.4 candidate status changes, per Gallagher-index point reduced.

5.2 Aggregate Party Correction and Territorial Redistribution

Six of the seven transfers are already required to reconcile the aggregate party totals produced by independent district PR with the canton-wide entitlements implemented by biproportionality. These six corrections must still be assigned to particular districts and are therefore territorial in their implementation. The seventh transfer does not further alter any party's canton-wide seat total. It is an additional territorial rearrangement needed to distribute the corrected party entitlements across districts while preserving every district magnitude.

All seven transfers are therefore consequences of biproportionality and all occur through changes in party–district cells. The distinction is analytical: six can already be inferred from aggregate party-seat correction, whereas one arises only from the two-dimensional distribution of those seats. In proportional terms, 85.7 percent of the transfers are required by aggregate party-total correction and 14.3 percent constitute additional territorial rearrangement beyond that minimum.

5.3 Candidate-Level Consequences of the Seven Transfers

Table 3 presents the seven transfers as paired candidate outcomes. Each row represents one parliamentary seat that moves between party lists within a district when the biproportional constraint is applied. Each transfer generates two candidate status changes: one candidate gains elected status and one loses it.

Table 3 Candidate occupants changed by biproportionality relative to independent district PR

District	Gains status under biproportionality	Loses status under biproportionality
Zurich IV	AL: Judith Stofer	Greens: Simon Meyer
Zurich VI	AL: Anne-Claude Hensch	FDP: Frank Rühli
Hinwil	FDP: Stephan Weber	SP: Advije Delihasani
Uster	GLP: Claudia Frei-Wyssen	Greens: Julian Andrea Croci
Winterthur-Land	EVP: Markus Schaaf	Greens: Ralf Hablützel
Andelfingen	The Centre: Konrad Langhart	SVP: Andrina Trachsel
Dielsdorf	FDP: Christian Müller	SVP: Christian Lucek

Note: Gains and losses refer to the observed biproportional allocation relative to independent district Webster allocation with the Zurich list-group quorum. Candidate rankings and candidate votes are identical under both allocations.

The table also illustrates why "candidate reversal" would be misleading terminology. None of these candidates reverses rank relative to another candidate. The number of seats to which the ranking applies changes. "Allocation-induced candidate status change" is therefore the more precise term.

5.4 Discordant Seat Allocations and Their Candidate-Level Incidence

The observed biproportional allocation contains two pairwise seat reversals in two of the eighteen districts. No discordant allocation occurs under independent district PR. In neither case is the locally strongest party overtaken; both reversals concern parties competing below the leading position.

In Hinwil, the SP receives 2,919 voter equivalents, or 13.9 percent, compared with 2,744 and 13.1 percent for the FDP. Independent district PR assigns two seats to the SP and one to the FDP. Biproportionality reverses their local seat ordering: the FDP receives two seats and the SP one. The transfer elects Stephan Weber of the FDP and removes elected status from Advije Delihasani of the SP.

In Winterthur-Land, the Greens receive 1,076 voter equivalents, or 7.46 percent, compared with 978 and 6.78 percent for the EVP. Independent district PR assigns one seat to the Greens and none to the EVP. Under biproportionality, the Greens receive no seat and the EVP one. Markus Schaaf of the EVP gains elected status, while Ralf Hablützel of the Greens loses it.

Table 4 Discordant seat allocations under biproportionality

District	Discordant allocation	Candidate-level consequence
Hinwil	SP: 13.9% and 1 seat; FDP: 13.1% and 2 seats.	Stephan Weber (FDP) gains elected status; Advije Delihasani (SP) loses elected status.
Winterthur-Land	Greens: 7.46% and 0 seats; EVP: 6.78% and 1 seat.	Markus Schaaf (EVP) gains elected status; Ralf Hablützel (Greens) loses elected status.

Note: The locally stronger party is listed first in each row. A discordant allocation occurs when that party receives fewer seats than the locally weaker party. Percentages indicate district vote shares.

Both reversals coincide directly with one of the seven gain–loss transfer pairs. They account for four of the fourteen candidate status changes. The remaining five transfers and ten status changes occur without reversing the local party ordering by votes and seats. For example, a party may lose a seat but remain ahead of a weaker competitor, or a smaller party may gain representation without overtaking the stronger party in seat totals.

This result establishes the relationship between the established and proposed concepts. Seat reversals identify a particularly visible subset of the local consequences of biproportionality. Allocation-induced candidate status changes capture the broader candidate-level incidence of all party–district threshold shifts.

5.5 Proximity to Local District Seat Boundaries

The affected party–district cells are close to district allocation boundaries under independent Webster allocation. Among the fourteen changed cells, the minimum relative quotient distance to gaining the next seat or losing the final awarded seat ranges from approximately 1.2 to 9.5 percent. The global constraint therefore modifies cells that are locally contestable rather than far removed from the final district seat boundary.

This finding is descriptive rather than explanatory. Many cells may lie close to a district threshold without being adjusted because the biproportional matrix is determined jointly by all party and district totals. The Zurich case contains only seven transfers, and gain and loss cells within the same matrix are not statistically independent. A cell-level regression would therefore overstate the available information. The appropriate conclusion is narrower: the affected cells are locally close to an alternative Webster allocation, but their selection for adjustment depends on the full two-dimensional matrix.

5.6 Vote Gaps across the Shifted Candidate Thresholds

The vote gaps across the shifted candidate thresholds reveal substantial heterogeneity. Five of the fourteen affected candidates are associated with a list moving between zero and one seat. In these cases, one allocation rule has no interior last-elected/first-non-elected threshold, so a conventional candidate-threshold margin is undefined. These transitions create or remove a party's local representation and are therefore reported separately.

For the remaining nine affected thresholds, the normalized vote gap between the relevant adjacent candidates ranges from 0.67 to 42.8 percent of mean candidate votes on the list, with a median of 20.1 percent. This quantity is not a percentage-point difference in candidate vote shares. A value of 42.8 percent means that the absolute vote gap between the candidates on either side of the shifted threshold equals 42.8 percent of the average candidate-vote total on that party–district list. The narrowest case is in Dielsdorf, where Christian Lucek is separated from the adjacent candidate by 40 votes. Other changes cross gaps of several hundred or more than one thousand votes.

The threshold gap does not explain why the candidate's status changes; the party–district entitlement does. Nor does a wide gap mean that a lower-ranked candidate defeats a more popular co-partisan. Candidate rankings remain unchanged under both allocation rules, and every higher-ranked candidate is elected before a lower-ranked candidate. The measure instead characterizes the point in the ranking crossed by the allocation change. Some status changes separate nearly tied adjacent candidates, while others extend or contract the elected set across a clearly differentiated ranking boundary.

6 Perturbation Benchmark and Robustness

6.1 Perturbing the Zurich Vote Matrix

Table 5 reports the perturbation results for seat transfers and aggregate proportionality. Across all three concentration parameters, the simulated elections produce a median of eight seat transfers and a 5–95 percent interval of five to twelve. The empirical value of seven lies between the 34th and 40th percentiles. Zurich 2023 therefore does not represent an unusually extensive disruption of local seat continuity under the calibrated perturbations.

The proportionality improvement occupies a different position in the simulated distributions. The observed Gallagher reduction of 0.91 lies between approximately the 8th and 20th percentiles, depending on the concentration parameter. Simulated elections generally produce larger proportionality gains from replacing independent district PR with biproportionality. Zurich 2023 thus combines an ordinary number of local seat transfers with a comparatively modest improvement in aggregate proportionality.

Table 5 Perturbation benchmark for transfers and proportionality

Dirichlet concentration	Outcome	Mean	5–95%	Empirical percentile
40	Seat transfers	8.12	5–12	39.8%
	Gallagher improvement	1.65	0.80–2.51	8.4%
80	Seat transfers	8.42	5–12	34.2%
	Gallagher improvement	1.65	0.74–2.59	10.2%
160	Seat transfers	8.17	5–12	38.3%
	Gallagher improvement	1.48	0.56–2.51	19.6%

Note: Each concentration condition contains 1,000 successful perturbations. The empirical values are seven seat transfers and a 0.91-point reduction in the Gallagher index. Gallagher improvement is calculated as the district-PR Gallagher index minus the biproportional Gallagher index.

Figure 2 shows the full distribution of seat transfers under each concentration parameter. The empirical value of seven lies slightly below the centre of all three distributions but well within their common range. The similarity of the distributions across the three perturbation conditions also indicates that the estimated extent of local reallocation is not highly sensitive to the selected concentration parameter.

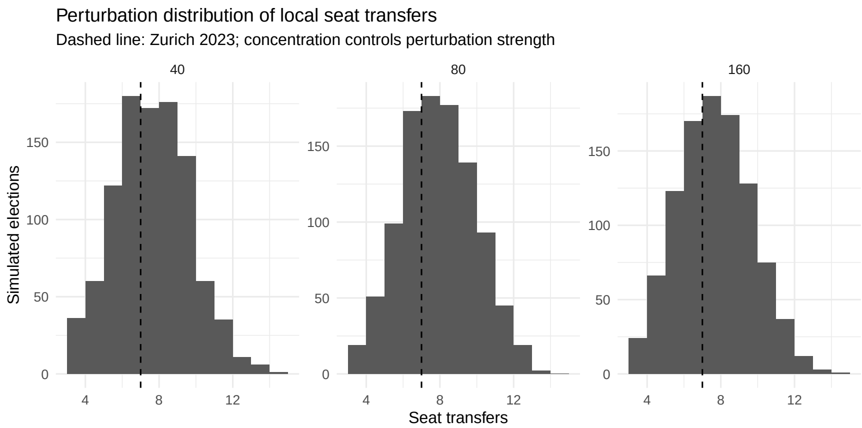

Fig. 2 Perturbation distributions of seat transfers. The dashed line marks the empirical Zurich 2023 value.

Figure 3 displays the relationship between the reduction in Gallagher disproportionality and the number of local seat transfers across the simulated elections. The positive association indicates that larger improvements in aggregate proportionality are generally accompanied by more extensive changes to the party–district seat matrix. The relationship is not deterministic, however: simulations with similar proportionality gains can involve different numbers of transfers, reflecting variation in how aggregate party corrections are distributed territorially. The Zurich 2023 result combines seven transfers with a Gallagher reduction of 0.91 and appears in the lower part of the simulated proportionality gains among elections with a similar number of transfers.

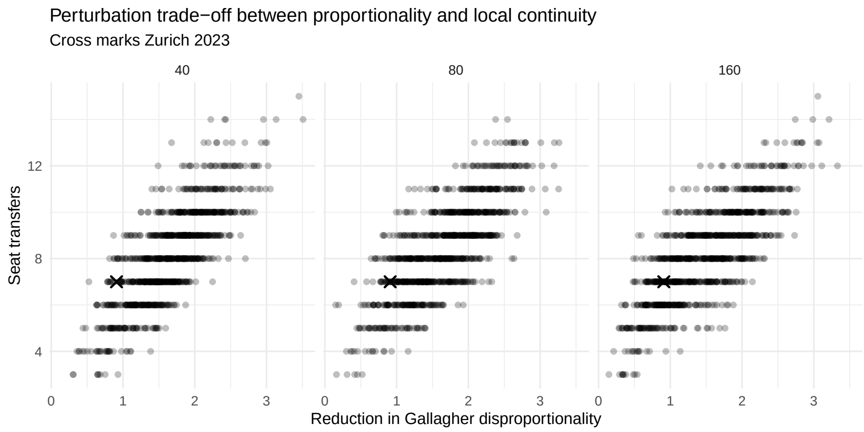

Fig. 3 Simulated relationship between improvements in Gallagher disproportionality and local seat transfers. Crosses mark Zurich 2023.

The seat-reversal benchmark reported in Table 6 yields a similar assessment of Zurich's empirical position. The observed two discordant districts lie between the 33rd and 41st percentiles of the simulated distributions. The observed two pairwise reversals lie between the 23rd and 29th percentiles. Zurich is therefore not unusually discordant under nearby vote configurations. Simulated elections typically produce three discordant districts and four pairwise reversals.

Table 6 Perturbation benchmark for discordant seat allocations

Dirichlet	Discordant districts			Pairwise seat reversals		
Concentration	Mean	5–95%	Empirical percentile	Mean	5–95%	Empirical percentile
40	3.09	1–6	38.1%	4.18	1–9	27.6%
80	3.16	1–6	33.5%	4.25	1–8	23.1%
160	2.91	1–5	41.4%	4.00	1–8	28.6%

Note: The empirical allocation contains two districts with discordance and two pairwise reversals. No district has a reversal involving the locally strongest party; the median simulated value for that outcome is also zero.

These simulations do not imply that seven transfers or two reversals should be expected in other cantons. The benchmark is conditioned on Zurich's district magnitudes, party contest structure, eligibility rule, and observed vote matrix. Its diagnostic value is narrower: the observed number of transfers and the incidence of seat reversals are not artefacts of the exact 2023 vote totals, and both remain within a stable range under moderate perturbations.

6.2 Counterfactual and Quorum Robustness

The primary result is unchanged when the quorum is removed: independent district Webster allocation still differs from the observed matrix by seven transfers. The legal list-group threshold is therefore not driving the primary Zurich result.

Applying an independent 5 percent threshold in every district produces twelve transfers, while applying the 5 percent district or 3 percent canton-wide condition separately to each party–district cell produces nine. The difference arises because the legal eligibility condition applies to the list group as a whole rather than independently to each local cell. This underscores the importance of modelling the institutional level at which electoral eligibility is determined.

Changing the allocation formula has a larger effect. Hare allocation produces five transfers, Webster seven, and D'Hondt twenty. This variation is not surprising: the formula changes party entitlements within every district and therefore alters more than the presence or absence of the biproportional constraint. It reinforces the choice of Webster as the primary counterfactual because it holds the divisor family constant. Alternative formulas demonstrate design sensitivity but do not represent equally clean estimates of the upper-apportionment effect.

Table 7 Robustness of local seat reallocation

Counterfactual specification	Changed cells	Seat transfers
Webster, Zurich list-group quorum	14	7
Webster, no quorum	14	7
Webster, independent 5% district threshold	24	12
Webster, cell-wise 5% or 3% rule	18	9
D'Hondt, Zurich list-group quorum	38	20
Hare, Zurich list-group quorum	10	5

Note: Changed cells report the L_1 distance between the observed biproportional seat matrix and the respective counterfactual allocation. Seat transfers equal one half of the number of changed seats across party–district cells.

7 Discussion

The Zurich results connect two levels of research that have largely remained separate. The apportionment literature has identified discordant seat allocations as a local consequence of reconciling district structures with global party proportionality (Maier et al. 2010). Research on personalized PR, by contrast, emphasises how candidate votes determine which individuals fill the seats assigned to a party. The present analysis shows how the two processes interact: changes in party–district entitlements alter the number of positions in the local candidate ranking that lead to election.

The distinction between seat reversals and candidate status changes is empirically consequential. Only two of the seven Zurich transfers produce classic pairwise reversals. In Hinwil and Winterthur-Land, a locally weaker party receives more seats than a stronger party, and four candidate statuses change as a direct result. Yet ten further candidate status changes occur in districts without any reversal of the local party ordering. The candidate-level incidence of global proportionality is therefore substantially broader than the most visible form of party-level discordance.

This finding refines the theoretical meaning of seat reversals. They are not a general synonym for every seat shifted by biproportionality. They identify cases in which the vote and seat rankings of two parties directly conflict. A cross-rule cell change asks a different question: whether the same party receives a different local entitlement under biproportionality and independent district PR. Candidate status changes arise from the latter. Treating the concepts separately avoids both understating candidate effects by counting only reversals and overstating reversals by labelling every transfer as one.

The sequential mechanism also clarifies the role of candidate votes. Biproportionality acts on the party–district entitlement, not on the ordering of candidates. Once that entitlement changes, the electoral threshold moves within a ranking that local voters have already determined. Candidate votes identify the individual whose status changes and the vote gap crossed by the new threshold; they do not cause the party to gain or lose the seat.

The threshold gaps nevertheless add substantive information. Christian Lucek's relevant boundary in Dielsdorf is separated by only 40 votes, while other status changes cross gaps of several hundred or more than one thousand votes. Five affected candidates are connected to transitions between zero and one party seat, for which no interior threshold exists under one of the rules. Mechanically identical one-seat changes can therefore have different candidate-level profiles.

The results also clarify the proportionality–continuity trade-off. Biproportionality reduces the Gallagher index by 0.91 points and the Loosemore–Hanby index by 2.25 points, while seven parliamentary seats acquire different occupants. Six transfers are already required by the correction of aggregate party totals, and one additional transfer arises from distributing the corrected totals across districts. All seven are territorial in implementation and all are consequences of the biproportional matrix.

These quantities do not establish that one allocation is normatively superior. Biproportionality is designed to treat party votes more equally across districts, and the observed improvement in aggregate proportionality is real. Independent district PR provides greater local allocation autonomy but produces party totals that deviate further from canton-wide vote shares. The empirical contribution is to make the exchange visible rather than to assign a single normative weight to its components.

The perturbation analysis provides context. Zurich 2023 is not unusually disruptive under nearby vote configurations. Seven transfers lie slightly below the simulated median, while the two discordant districts and two pairwise reversals also fall below simulated averages. The proportionality improvement is comparatively modest. The case should therefore not be presented as an exceptional failure of biproportionality. It demonstrates an ordinary structural mechanism whose local incidence and aggregate payoff vary across elections.

The findings have implications for transparency. Official reporting generally presents the final seat matrix and the elected candidates but does not always make the relationship between upper apportionment, district entitlements, and candidate thresholds readily visible. Reporting discordant allocations is useful because they are intuitively striking, but it is insufficient. Five of seven transfers in Zurich change candidate outcomes without producing a classic reversal. A fuller explanation would identify all party–district cells changed by the lower apportionment and show which candidate rank becomes decisive as a result.

The results may also matter for campaign incentives, although the present design does not estimate them. Candidates face a two-level electoral problem. Personal votes cast in the district determine their position within the local list, so incentives for localized candidate-centered campaigning remain intact. Yet the electoral value of any rank depends on how many seats the party ultimately receives in the district, an entitlement constrained by the party's canton-wide performance and the lower apportionment. Candidates and local party organizations may, therefore, have reason to combine personal vote seeking with broader party mobilization and coordination beyond the district. This does not imply that candidates can substitute canton-wide campaigning for local support: they cannot receive personal votes outside their district, and the location of an additional party seat is not known in advance. The likely implication is a dual rather than a displaced incentive structure. Whether candidates perceive and respond to these incentives remains an empirical question.

The comparison with mixed-member systems remains useful at the level of the underlying institutional tension. Mixed-member systems combine territorial or personal representation with a separate mechanism for achieving overall party proportionality (Shugart and Wattenberg 2001; Bochsler 2023; Behnke 2025; Flis, Behnke, Lorenc, and Salamon 2025; Flis, Grofman, and Kaminski 2025). Biproportionality does so without separate tiers or mandate classes. Its local consequences arise through the seat matrix, and the candidate-level effects identified here extend the study of compensatory representation to a formally uniform electoral system.

Several limitations remain. First, the analysis concerns one canton and one election. It demonstrates and quantifies a mechanism but cannot estimate an average effect across biproportional systems. A multi-election analysis across Swiss cantons would provide stronger leverage on when seat reversals and matrix distance become large and whether district-magnitude inequality, party fragmentation, or geographic concentration predict them.

Second, the counterfactual is mechanical. It holds votes and candidate rankings constant and therefore isolates the allocation rule. It does not represent a complete alternative electoral equilibrium. This is appropriate for identifying immediate allocation consequences but restricts claims about what would occur after actual institutional reform.

Third, the local quotient distances and candidate-threshold gaps are descriptive. Seven paired transfers and two seat reversals provide insufficient independent variation for multivariable explanatory models, and the cells are jointly determined by the matrix. Comparative data would be needed to explain systematically which locally close cells are selected for adjustment and when those changes generate discordance.

Fourth, the perturbation model is calibrated to Zurich 2023. It tests local stability rather than institutional generality. A broader factorial simulation would need to vary the number of districts, district-magnitude inequality, party fragmentation, geographic concentration, eligibility thresholds, and contest structure.

Finally, the study establishes local allocation conditionality rather than behavioral accountability effects. The normative distinction between ordinal personalization and stronger local control is derived from the institutional design, not from observed public reactions. Whether voters understand seat reversals and candidate threshold shifts, whether wide-gap changes are regarded as less legitimate than near-ties, whether candidates adapt their campaign strategies, and whether affected outcomes influence perceived legitimacy requires survey, experimental, or longitudinal evidence.

8 Conclusion

Biproportional representation has local consequences at both the party and candidate levels. Existing research captures the most visible party-level consequence through the concept of discordant seat allocations or seat reversals. This article shows that the same globally compensatory logic also changes candidate outcomes in cases that do not produce a classic reversal.

In Zurich 2023, the official biproportional matrix differs from independent district Webster allocation in fourteen of 188 party–district cells. These differences represent seven seat transfers, fourteen candidate status changes, and different occupants in seven of the parliament's 180 seats. Two transfers generate pairwise seat reversals: the FDP overtakes the locally stronger SP in Hinwil, and the EVP overtakes the locally stronger Greens in Winterthur-Land. No reversal occurs under independent district PR. The remaining five transfers nevertheless change ten candidate statuses without reversing the local party ordering.

This distinction is the central conceptual result. Seat reversals are a particularly strong form of party-level discordance, but they are not a complete measure of the local incidence of global proportionality. Candidate outcomes depend on every change in a party's district entitlement. In personalized list systems, the number of seats assigned to the party determines how far down the locally voter-shaped ranking mandates extend.

The candidate-threshold vote gap provides a secondary description of these effects rather than an explanation of them. Some allocation changes cross almost tied adjacent candidates, others cross clearly differentiated points in the ranking, and some move a party between zero and one seat. In every case, the voter-determined order remains intact. What changes is the number of ranks that result in election. Wide-gap threshold shifts make this conditionality particularly visible: personalized voting controls the order of candidates, while the biproportional matrix controls the number of available local mandates.

Biproportionality improves aggregate proportionality in Zurich, reducing the Gallagher index from 2.64 to 1.73 and the Loosemore–Hanby index from 4.81 to 2.55. The perturbation benchmark indicates that the observed seven transfers and two seat reversals are not unusually numerous under Zurich-like vote configurations, while the proportionality gain is comparatively modest. The broader lesson is not that biproportionality generally weakens candidate choice. It is that global proportionality has a measurable local incidence that extends from party-level seat reversals to the personal composition of parliament.

A Formal Decomposition of Seat and Candidate Changes

Let s^B_{pd} denote the number of seats allocated to party p in district d under biproportionality and let s^D_{pd} denote the corresponding allocation under independent district PR. The cell-level difference is

$$\Delta_{pd} = s^B_{pd} - s^D_{pd}. \tag{2}$$

Because both allocations distribute the same total number of seats, positive and negative differences balance. The total number of seats transferred between party–district cells is therefore

$$T = \frac{1}{2}\sum_{p,d} \left|s^B_{pd} - s^D_{pd}\right|. \tag{3}$$

If candidate rankings are held fixed and each party–district list contains enough candidates to fill the larger of the two seat entitlements, the number of candidates whose election status changes is

$$C = \sum_{p,d} \left|s^B_{pd} - s^D_{pd}\right| = 2T. \tag{4}$$

To distinguish the correction of aggregate party totals from the additional local reallocation required by the district constraints, define the aggregate seat total of party p under allocation rule R as

$$q^R_p = \sum_d s^R_{pd}. \tag{5}$$

The minimum number of unit seat transfers required to transform the aggregate party totals under independent district PR into those under biproportionality is

$$T_P = \frac{1}{2}\sum_p \left|q^B_p - q^D_p\right|. \tag{6}$$

Because aggregation over districts cannot increase the L_1 distance, $T_P \leq T$. The excess number of local transfers relative to this aggregate minimum is therefore

$$T_L = T - T_P. \tag{7}$$

Here, T_L does not identify transfers that individually leave aggregate party totals unchanged. Rather, it measures the additional unit reallocations required to implement the aggregate party correction across districts while preserving every district magnitude.

In the Zurich 2023 application, $T = 7$, $T_P = 6$, and $T_L = 1$. The aggregate differences in party totals require at least six transfers, whereas the observed party–district matrix requires seven. The excess transfer reflects an indirect territorial routing of one aggregate correction. In particular, the transfers from the SVP to the FDP in Dielsdorf and from the FDP to the AL in Zurich VI form the two-step path SVP–FDP–AL, while their aggregate net effect is a one-seat transfer from the SVP to the AL.

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Competing interests

The author declares that he has no relevant financial or non-financial interests to disclose.

Data availability

The underlying data are publicly available through the Canton of Zurich data catalogue under *Resultate und Wahlbeteiligung Kantonsratswahl 2023* (Amt für Statistik und Daten des Kantons Zürich 2023). The analysis uses the canton-level list results (KTZH_00002183_00004285), district-level list results (KTZH_00002183_00004286), and district-level candidate results (KTZH_00002183_00004288). The processed data and replication materials are available in an anonymized OSF repository.

Code availability

The R code used for the counterfactual allocation, validation, robustness analyses, and perturbation simulations is available in the OSF repository.

Ethics approval

Not applicable. The study does not involve human participants, surveys, interviews, experiments, or non-public individual-level data.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Use of generative AI and AI-assisted technologies

The author used AI-based tools for coding assistance and language editing. The author reviewed and verified the code, analyses, and manuscript content and takes full responsibility for the final version.

Author contributions

The author is solely responsible for the conception and design of the study, data collection, analysis, interpretation, and writing of the manuscript.

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